AVO Carbon KUNSHAN

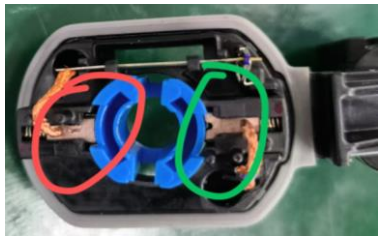
8D REPORT
8D 报告

表单编号：AVO-QP-23-01/A

Title:标题:	Complaint No.:2025.7.9	Date Opened: 开启日期:	Last Updated: 最近更新:	Days	Evidence
U611刷架组件碳刷混料 U611 brushcard assembly carbon brushes mixed material		2025.7.9	2025.7.13		

Product/Process Information:产品/过程信息:	Organization Information:组织信息:
NAT02-U611	AVO Carbon Anhui

D1 Team: D1：组建团队	Name 名字	Dept 部门	Phone 电话
Champion(Problem solving experts)			
Team Leader:团队领导:	coby.liu	QE	13170063212
Team Members:团队成员:	heqin	quality Supervisor	13306265707
	wangjiaxue	production Supervisor	13328055803
	chuqiming	Mechanical repair-Supervisor	18126738790
	zhawujiu	production Supervisor	13326954233



D2 Problem Statement: D2 问题陈述:	
Part No.:	k0922003
Part name:	NAT02-U611
Date of Detection	2025.6.19
Location of Detection	DY客户现场
defect quantity	1 PCS
Quantity manufactures	19700
Defect rate	0.01%
problem description	碳刷错装不良
Failure effect at customer production line	此缺陷在客户现场发现并反馈，目前发现1pcs
Failure effect at OEM（OKM）	电机测试中可以识别不良，低风险
Fialure effect on final customer（field）	电机测试中可以识别不良，低风险
How often does the problem occur?	个例
What is the problem history	几年前有发生过
Other customer parts potentially affected?	NO
Preliminary risk assessment	无流出风险

End of D2：2025.7.9

12小时内

D2 - Failure Visualization'!A

D3 Interim Containment Action(s):D3 制定临时措施/围堵：（详见附页）

对在制品、在库品进行全部分选：在库12050pcs，在制点焊组件231pcs . --何秦、查五九 2025.7.9

Labeling after complaint () / Acc. to specific request applicable to this complaint. How? Which shipments? /

对全检排查标签进行标记“碳刷OK” --何秦 2025.7.10

Are all affected areas and departments informed? /If yes, please specify. If no, why not?/

不良缺陷对相关工序人员已传递 工序人员：碳刷点焊、终检、GP12、生产领班、主管；质量巡检、领班、机修领班 2025.7.9

Description of Firewall (until verification of actions in D6) /refer to the list of containment actions above/

增加GP12检验工序，全检碳刷是否错装 工序：终检、Gp12 2025.7.9

Agreement of the customer on process or product-changing containment actions.

目视检碳刷符合客户检查标准

24小时内

Quality Alerts

D4 Root Cause(s) and Escape Point(s): D4 确定根本原因和问题逃逸点:

Occurrence（为何不良会发生？）	
Technical root cause-TRC	点焊作业区域换线清线不完全
Managerial Root Cause-MRC	点焊区域清换线标准未有形成文件，在FMEA中未识别出该区域的混碳刷的风险。
Non-Detection（为何不良会逃逸）	

D4 - Ishikawa'!A1

Technical root cause-TRC	过程检验“碳刷有无错装”动作检验标准中未明确。
Managerial Root Cause-MRC	在FMEA中未识别出“检验标准中未规定检碳刷错装的动作”存在流出风险。
Technical root cause-TRC	CCD识别图像时受条件影响。
Managerial Root Cause-MRC	在FMEA中未识别出因载具位移变化的带来的失效风险，未模拟训练此类异常（缺乏相关设备的运行经验，或未收到相关影响现场测试数据）

Causing Process: /Explanation of the causing process

Details refer D4-Ishikawa and D4-5Why

Why don't defects occur 100%? 碳刷点焊前或换线，未能对作业区域进行清线确认到位，出现个例情况的混用。

Occurrence: Does the analysis prove that no other root cause can be linked and that the failure rate is coherent with baseline? /YES or NO/ Yes.

If No,why not:

Detection: Proof that the test coverage is sufficient. /YES or NO/ : Yes

If no, Please specify:

二个工作日

Reproduction of the defect possible?	Yes
If Yes,Result:	1. 点焊区已点焊的碳刷增于325PCS组件，目视100%检，未发现异常； 20525.7.9 丁文涛
	2. 组装区已点焊的碳刷增于231PCS组件，目视100%检，未发现异常； 20525.7.9 查五九
	3. 12050PCS已完成的产品，目视100%检，未发现异常； 2025.7.10 何秦
	4. 查阅2025.6~2025.7 NATA02 产品过程记录及终检、GP12检查记录，共26批次，未发现有混料异常。 2025.7.9 coby 、王佳雪
If no, Why not:	NA

Risk assessment (Probability) before implementation corrective actions:

1.点焊前或切换型号作业清线要求培训 2.对其他类型相似产品的同步开展作业前清线点检自查，领班、巡检复检确认。 王佳雪 、coby 2025.7.9

2 利用不良电刷样本重验证CCD识别，采用模拟载具位移变化反复验证。 储启明、coby 2025.7.9

Production period affected from:	From：NA To： NA
Expected number of further non-conformities:	0 PCS

D5 Chosen Corrective Action(s) and estimated effectiveness: D5 选择措施和有效性评估：

针对根本原因采取的行动(D4):			
Occurrence			
TRC1	明确点焊区域的清线点检区域，清线位置及内容。	Estimated effectiveness [%]：100%	Effective from：07-10
MRC1	更新清线&换型作业标准书，将点焊区域的换清型号点检要求完善到标准中	Estimated effectiveness [%]：100%	Effective from：07-10
Non-Detection			
TRC1	明确检验过程增加“碳刷有无错装”检验动作	Estimated effectiveness [%]：100%	Effective from：07-10
MRC1	完善FEMA失效模式，增加“碳刷有无错装”检验动作，并更新检验指导文件	Estimated effectiveness [%]：100%	Effective from：07-10
TRC2	识别出影响CCD识别图像影响的条件，模拟训练方式对CCD进行设定，验证并完成设置	Estimated effectiveness [%]：100%	Effective from：07-9
MRC2	完善FEMA失效模式，增加影响CCD识别图像影响的条件：比如载具晃动等	Estimated effectiveness [%]：100%	Effective from：07-9
	Completed on:7-10	Responsible：kerwin&coby&Jim&qimignchu	

D6 Implemented Corrective Action(s): D6 执行措施：

Occurrence					
TRC1	明确点焊区域的清线点检区域，清线位置及内容。	Effectiveness	100%	Responsible：coby&kerwin	
	Planned introduction on:	7/10	introduced on:	7/10	Effective from:7-10

14个工

D6-Implemented corrective

MRC1	更新清线&换型作业标准书，将点焊区域的换清型号点检要求完善到标准中		Effectiveness	100%	Responsible：kerwin	工作日（具体依客户要求）
	Planned introduction on:	7/10	introduced on:	7/10	Effective from:7-10	
MRC3	Update the PFMEA		Effectiveness	100%	Responsible：Ding Shaoyang	
	Planned introduction on:	1/20	introduced on:	1/18	Effective from:01-18	
Non-detection						
TRC1	明确检验过程增加“碳刷有无错装”检验动作		Effectiveness	100%	Responsible：coby	
	Planned introduction on:	7/10	introduced on:	7/10	Effective from:2025/7/10	
MRC1	完善FEMA失效模式，增加“碳刷有无错装”检验动作，并更新检验指导文件		Effectiveness	100%	Responsible：coby	
	Planned introduction on:	7/10	introduced on:	7/10	Effective from:2025/7/10	
TRC2	识别出影响CCD识别图像影响的条件，模拟训练方式对CCD进行设定，验证并完成设置		Effectiveness	100%	Responsible：JIM&QIMINGCHU	
	Planned introduction on:	7/9	introduced on:	7/9	Effective from:7-9	
MRC2	完善FEMA失效模式，增加影响CCD识别图像影响的条件：比如载具晃动等		Effectiveness	100%	Responsible：Coby	
	Planned introduction on:	7/9	introduced on:	7/10	Effective from:7-11	

Removal of containment actions:

Responsible：

verification:

Responsible：

D7 Prevent Actions: D7 预防问题重现（更新的文件，图纸，PQP，WI etc.）： 二个工作日

1. 更新清线&换型作业标准书，将点焊区域的换清型号点检要求完善到标准中	Responsible：kerwin	Schedule: 07-15	completed on：	工装图纸!A1
2. 完善FEMA失效模式，增加“碳刷有无错装”检验动作，并更新检验指导文件	Responsible:coby	Schedule: 07-15	completed on：	
3. 识别出影响CCD识别图像影响的条件，模拟训练方式对CCD进行设定，验证并完成设置	Responsible:JIM	Schedule: 07-9	completed on： 7-9	
4. 完善FEMA失效模式，增加影响CCD识别图像影响的条件：比如载具晃动等	Responsible:coby	Schedule: 07-15	completed on：	
5. 质量通报及相关培训记录	Responsible:heqin&wangjiaxue	Schedule: 07-10	completed on： 07-10	

经验教训转移到其他流程、产品和工厂Lessons Learned transfer to other Processes, Products, Plants：

Could the root cause affect other processes, products or sites?	Yes
If yes: Which other processes, products or sites may be concerned?	NAT02-V
If yes: Which departments do you inform (e.g. Lessons Learned Coordinators)?	生产部、质量部、维修部

If no: Why don't you expect other processes, products or plants to be concerned? LLC!A1

D8 Final decision / D8 结论: Until 6th July this kind of part was deliveried 100800PCS(total more than 5 batches) and no concentricity was found,It can be closed. 一个工作日

Participants/参加者： Ding shaoyang Zhu zhuhua Allan Regiel Steven Li Zhoushaoyu Sam Wan Sunny Liu

Accomplished on/完成日期：

Decision on complaint//结论: ☑ 接受Accepted ☐ 拒绝Rejected

Plant manager： Name： Date: Signature:

Champion/Expert： Name： Date: Signature:

Team Leader:	Name:	Date:	Signature:		